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1 Introduction

1.1 Purpose

The purpose of creating 'SyncSphere' is to provide a comprehensive, feature-rich communication platform that enables seamless digital interaction. By offering robust server creation, real-time communication channels, and advanced social features, the app aims to create a dynamic virtual community space. 'SyncSphere' combines innovative communication tools such as voice and video channels, server management, and user interaction capabilities to help users connect, collaborate, and communicate effectively.

1.2 Document Conventions

The higher level requirements are written in bold letters. The App name wherever used is enclosed between ' ' and is in bold letters. The main headings are written in Arial fonts of size 14 and the detailed content is written in Arial fonts of size 12.

1.3 Intended Audience and Reading Suggestions

This SRS document can be read by the Design team while preparing Design Models and Prototype Model. The Design team must first look at the Product Functions, User characteristics, and classes with the user documentation and then move toward the detailed requirements specified in the document. It can also be used by the development team to identify the requirements correctly. The developer must look at the Classes and characteristics, also consider the hardware and software interfaces of the system and then, according to the requirements, start coding the system. This SRS is used in testing while identifying various test cases of the system. The tester must first see the assumptions and dependencies and according to the requirements design the test cases for the system.

1.4 Product Scope

The 'SyncSphere' webApp will include user authentication, server creation, social interaction features, voice channels, video meet, server management, live chat, media sharing, theme customization, and member management systems. This document outlines the functionalities and requirements necessary for the development of the webApp.

1.5 References

- **IEEE Template for System Requirement Specification Documents**:
  `https://goo.gl/nsUFwy`
- **Websites referred for learning**:
  `https://nextjs.org/docs`
  `https://socket.io/docs/v4/tutorial/introduction`
  `https://www.prisma.io/docs/getting-started`
- **Similar App for reference**:
  `https://discord.com/`

2 Overall Description

2.1 Product Perspective

SyncSphere is a communication platform designed to enable seamless interaction between individuals and groups through text messaging, voice calls, video conferencing, and file sharing. It will allow the creation of customizable virtual spaces, where individuals and groups can gather to collaborate, share content, and engage in discussions on various topics. The platform aims to provide an intuitive interface for organizing conversations into distinct channels, supporting multimedia sharing, and fostering dynamic, interactive communities.

2.2 Product Functions

User Features

- **User Authentication**
  - User Registration and Login: Allow users to create accounts and log in using email or mobile number. Implement secure authentication mechanisms (e.g., OAuth, two-factor authentication).
  - Profile Customization: Users can set up their profiles with details such as username, avatar.
- **Server and Community Management**
  - Server Creation: Users can create and personalize servers with a unique name, icon, and description.
  - Role-Based Permissions: Server admins can create roles with specific permissions (e.g., managing channels, kicking users, mark users as moderator) and assign them to members.
  - Invite System: Users can generate unique invite links or codes to allow others to join their servers.
- **Communication Features**

  - Text Channels: Users can send messages in text channels, with support for images, and file attachments.
  - Voice Channels: Real-time voice communication in dedicated channels with options to mute or adjust individual volume levels.
  - Video Calls: Support for one-on-one and group video calls, along with screen sharing.
  - Direct Messaging (DMs): Private messaging between individual users, with options for group chats.
  - Channel Management: Create multiple text and voice channels within a server, each configurable with permissions and topics.

- **Media and File Sharing**
  - File Uploads: Allow users to share files, images, and videos with configurable file size limits.

- **Search and Discovery**
  - Global Search: Allow users to search for servers, channels.

- **Customization & Security**
  - Themes: Users can switch between light and dark mode or customize themes if enabled.
  - Two-Factor Authentication (2FA): Enhance account security with 2FA.

Administrative Features (Admin)

- **Server Management**
  - User Management: View, kick or mark moderator users from the server.
  - Role Assignment: Assign roles and permissions to members to control access and responsibilities.
  - Channel Configuration: Manage and edit channel permissions, topics, and visibility.

- **Customization**
  - Server Customization: Add server banners, icons, and names to match the community's branding.

2.3 User Classes and Characteristics

SyncSphere for Regular Users
'SyncSphere' offers users a comprehensive digital communication experience tailored to their social and collaborative needs. As a platform user, individuals can easily register and log in using email or social media accounts. Users can create and manage their profiles, customize personal settings, and join multiple servers based on their interests. The app provides robust communication tools including text, voice, and video channels, enabling seamless interaction across different modes. Regular users can create their own servers, invite members, manage roles, and engage in various communication formats. Social features allow users to follow other members, share media, and participate in community discussions. Progress tracking of server interactions, achievement badges, and customizable notification settings help users stay connected and engaged.

SyncSphere for Administrators/Server Owners
'SyncSphere' provides Administrators and Server Owners with comprehensive tools to manage digital communities effectively. They have advanced permissions to create, configure, and moderate servers, controlling user access and implementing strict role-based management. Administrators can customize server appearance, set communication rules, manage member permissions, and moderate content to ensure a safe and productive environment. They have access to detailed analytics about server engagement, user interactions, and platform performance. The admin toolkit includes advanced moderation features like message deletion, user warnings, and temporary/permanent banning capabilities. They can send targeted notifications, manage server invitations, and configure security settings to protect their community's integrity.

2.4 Operating Environment

Supported Devices for 'SyncSphere'

- Supported Platforms:
  - Web Application (Primary Platform)
  - Desktop Browsers: Chrome, Firefox, Safari, Edge
  - Mobile Browser Compatibility: Chrome Mobile, Safari Mobile
- Minimum System Requirements:

  - Processor: Intel Core i3 (7th Gen) or equivalent, AMD Ryzen 3
  - RAM: Minimum: 2 GB, Recommended: 4 GB
- Browser Requirements:
  - Modern browser with WebRTC support
  - JavaScript enabled
  - WebSocket compatibility
- Network Connectivity:
  - Wi-Fi or cellular data
  - Minimum 10 Mbps download speed
  - Stable internet connection for voice/video features
- Storage:
  - Browser cache: 900 MB
  - No local storage installation required
- Recommended Browser Versions:
  - Chrome: Latest 2 versions
  - Firefox: Latest 2 versions
  - Safari: Latest version
  - Edge: Latest version
- Distribution:
  - Web-based (No app store distribution)
  - Accessible via web browser

2.5 Design and Implementation Constraints

1. **Technology Stack for 'SyncSphere'**
   - Frontend: Next.js (React Framework)
   - Backend: Next.js API Routes
   - Authentication: Clerk
   - Database: Prisma ORM with MySQL
   - Real-time Communication: Socket.io
   - File Storage: UploadThing
   - State Management: React Hooks
   - Styling: Tailwind CSS
2. **Performance Constraints**
   - Response Time:less than 300ms for critical operations
   - Scalability: Horizontal scaling support
   - WebSocket Performance: Low-latency real-time updates
   - Concurrent User Handling: Minimum 10,000 simultaneous connections
3. **Technology Considerations**
   - Fully web-based platform
   - Responsive design for multiple screen sizes
   - Cross-browser compatibility
   - WebRTC support for video/voice channels
   - Secure, encrypted communication channels
4. **Time Constraints**
   - Development timeline: Complete project delivery by first week of Feb 2025

2.6 User Documentation

User documents such as the About Us page, FAQs, and quick tips on how to use this application will be provided and available on the app with the deployment of the product.

2.7 Assumptions and Dependencies

Assumptions:

- **Tech-literate users:** We assume most users will have basic computer and internet literacy to navigate the application effectively.
- **Reliable internet access:** Consistent and stable internet connections are necessary for all users to access the platform.
- **Accurate data entry:** We assume users will provide accurate and complete information during registration and profile updates.
- **Third-party services:** We may rely on third-party services for payment processing, insurance verification, or other functionalities.

Dependencies:

- **Secure cloud hosting platform:** A reliable and secure cloud platform is required for hosting the application and ensuring data security.
- **Database management system:** A robust database management system is needed to store and manage sensitive patient data.
- **Email and notification services:** Reliable email and notification services are crucial for appointment reminders, patient communication, and system updates.

3 External Interface Requirements

3.1 Hardware Interfaces

The minimum hardware requirements for Medconnect are 500 Megahertz CPU and 128 megabytes of RAM. A system with these specifications can handle a network of approximately 1000 edges and nodes. For bigger networks, additional memory is required.

3.2 Software Interfaces

The website FitnessFlow needs to connect to the MongoDB database to fetch the details of users dynamically. The machine needs support for Dart/Flutter.

3.3 Communication Interfaces

The App requires an internet connection to communicate with the server and to get all the updates.

4 System Features

4.1 User Authentication

Functional Requirement: User Management
Description: Allows users to register, log in, customize profiles, recover passwords, and manage accounts.

Actors:

- End User
- System Administrator

Input:

- Registration: Username, email, password, profile details, or OAuth.
- Login: Email/username, password.
- Password Recovery: Email for verification.
- Profile Update: Updated profile details.

Output:

- Successful account creation, login, or recovery confirmation.
- Error messages for invalid inputs or failed actions.

Pre-condition:

- The user must have a device with internet access.
- The system database must be active and accessible.

Post-condition:

- User account is created, updated, or recovered.
- User gains access to the platform.

Exception:

- Registration: Duplicate email/username.
- Login: Incorrect credentials or account lockout.
- Password Recovery: Invalid email or expired recovery token.

4.2 Server Management

Description: Enables users to create, edit, and delete servers, and assign roles and permissions. **Actors:**

- Registered Users
- Server Administrators
- Moderators

Input:

- Server details (name, icon)
- Action type (create, edit, delete, add, remove, assign)

Output:

- Updated server list (created/edited/deleted server)

Pre-condition:

- The user is logged in and has the necessary permissions (e.g., admin or moderator) to perform server management actions.

Post-condition:

- The server is created, updated, or deleted.
- User memberships and roles are updated as specified.

Exception:

- Insufficient permissions to perform an action.
- Invalid input data (e.g., duplicate server name or invalid role assignment).

4.3 Channel Management

Description: Ability to create, organize, and manage text, voice, and video channels within servers. **Actors:**

- Server admins, moderators, and members with channel management permissions.

Input:

- Request to create, edit, delete, or configure a channel.
- Permissions and access settings for the channel.

Output:

- New, updated, or removed channel.
- Adjusted access permissions or configurations.

Pre-condition:

- User must have appropriate permissions to manage channels.
- Server must exist.

Post-condition:

- Channels are updated as requested with corresponding configurations.
- Permissions for users reflect the changes made.

Exception:

- Operation fails if the user lacks sufficient permissions.
- Operation fails if the server is inactive or exceeds the maximum channel limit.

4.4 Text-Based Communication

Description: Enable real-time messaging between users and within group channels, including multimedia sharing and message interactions. **Actors:**

- Registered users
- Server moderators/admins

Input:

- Text message, multimedia file from the user

Output:

- Delivered message or file visible to intended recipients

Pre-condition:

- User is logged in and has access to the server/channel
- Active internet connection

Post-condition:

- Message or file successfully delivered and stored in the channel/message history

Exception:

- Message delivery failure due to network issues or server downtime
- Unauthorized access to restricted channels, resulting in access denial

4.5 Voice and Video Communication

Description: The platform supports high-quality voice and video calls, including group calls, with features like screen sharing, mute/unmute, and video toggle. **Actors:**

- Registered users (call participants)

- Communication server

Input:

- Initiate call request (voice or video)
- Participants' audio/video streams
- Commands (e.g., mute/unmute, toggle video, end call)

Output:

- Real-time voice and video streams
- Call status (active, ended)

Pre-condition:

- Users must be logged in and have necessary permissions for the call.
- Devices must have functional microphones, cameras, and stable network connections.

Post-condition:

- Active call ends, and all resources are released.
- Call logs/records are updated if applicable.

Exception:

- Network disruption may cause call disconnection or degraded quality.
- Device hardware issues may prevent participation (e.g., microphone/camera not detected).
- Insufficient permissions or server errors may block call initiation.

4.6 Administrative Tools

Description: Administrative tools enable server owners and moderators to monitor activity, manage users, and enforce rules within servers and channels. **Actors:**

- Server Owners
- Moderators

Input:

- Commands or actions from server owners or moderators (delete messages, edit roles)

Output:

- Updated server/user status (remove user, modified permissions)

Pre-condition:

- The actor must have appropriate administrative permissions.
- The server must be active and accessible.

Post-condition:

- Changes to server settings, user permissions, or logs are successfully applied and saved.

Exception:

- Insufficient permissions to perform the requested action.

5 Other Nonfunctional Requirements

5.1 Performance Requirements

- **Response Time:** The system should provide a response time of less than 2 seconds for 95% of user interactions.
- **Scalability:** The system should be able to handle a minimum of 1000 concurrent users without significant performance degradation.
- **Throughput:** The system should be capable of handling at least 100 messages per second across all channels during peak times.

5.2 Availability and Reliability

- **Uptime:** The application should have a 99.9% uptime, ensuring minimal downtime for maintenance.

5.3 Security Requirements

- **Authentication:** The system should require secure authentication mechanisms, such as email/password, two-factor authentication (2FA), or OAuth for login.
- **Authorization:** User roles (e.g., Admin, Moderator, Member) should be clearly defined, and users should have permissions only for the actions permitted by their role.
- **Audit Logs:** All user actions (e.g., login attempts) should be logged for auditing purposes.

5.4 Usability Requirements

- **User Interface:** The user interface should be intuitive, responsive, and easy to navigate on both desktop and mobile platforms.
- **Accessibility:** The application should comply with WCAG (Web Content Accessibility Guidelines) 2.1, ensuring accessibility for users with disabilities.

5.5 Maintainability and Extensibility

- **Modular Design:** The system should be designed with modular components so that new features can be added or existing features can be modified without affecting other parts of the system.
- **Code Documentation:** The source code should be well-documented to ensure ease of understanding and maintenance.
- **Version Control:** The codebase should use a version control system (e.g., Git) for proper tracking of changes, collaboration, and rollback when needed.

5.6 Network Requirements

- **Bandwidth:** The system should be able to work efficiently on networks with at least 1 Mbps bandwidth, ensuring stable communication and media sharing.